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LAB REPORT
(QUICK SORTING)

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Machine Problem Description: Faulty Machine Finder in a Factory

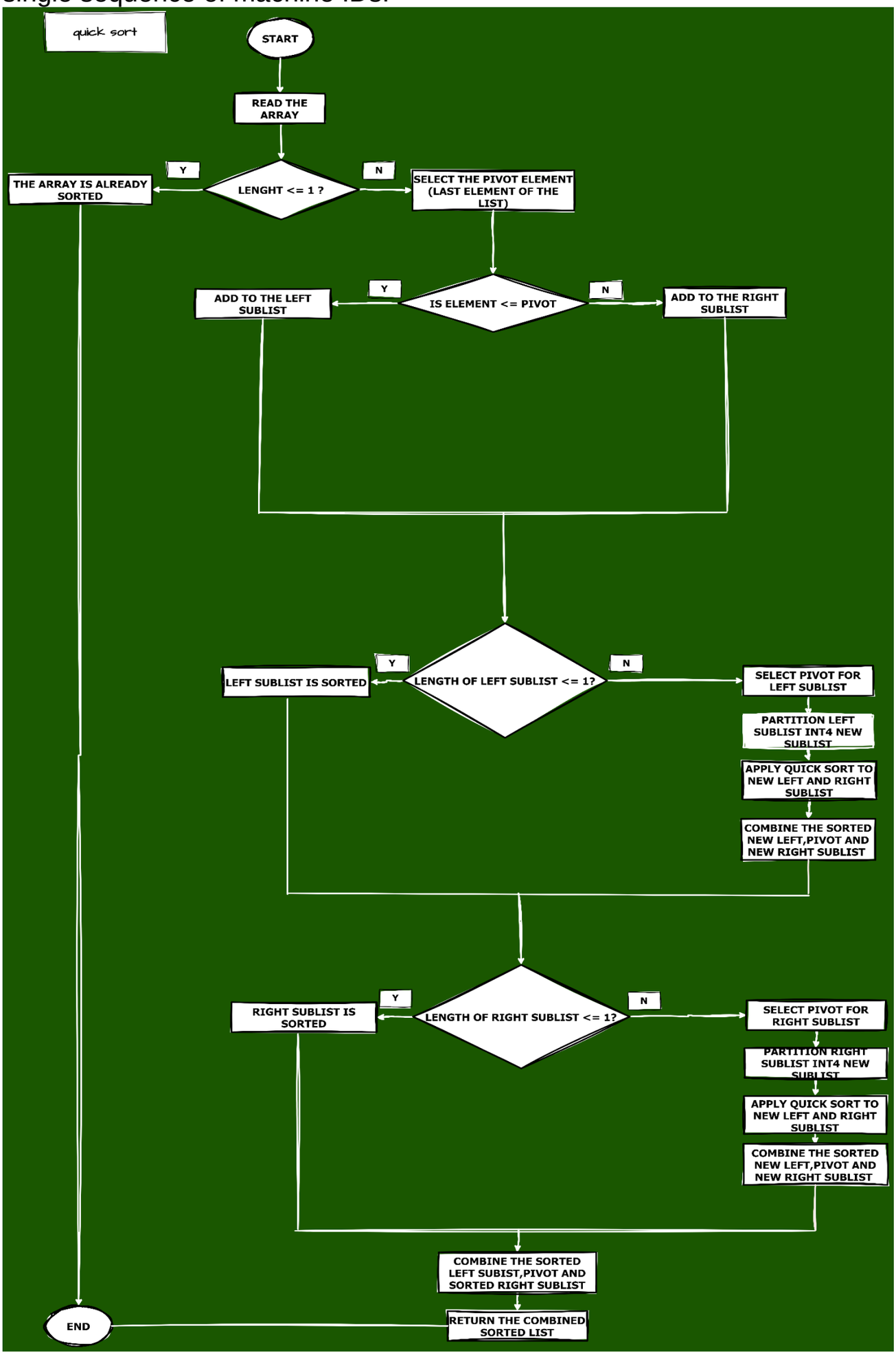
In a factory, multiple machines are producing items. Each machine is identified by a unique ID, and most machines produce an even number of items, meaning their IDs appear twice in the list of machines. However, one machine is faulty and produces an odd number of items, which means that the faulty machine's ID appears only once in the list. The goal is to sort the list of machine IDs using different sorting algorithms and then identify the faulty machine by applying the **XOR operation** to find the ID that appears only once.

QUICK SORT:

It employs a divide-and-conquer approach to sort the list. The algorithm selects a pivot element and divides the array around the pivot and recursively sorts sub-arrays on both sides of the pivot.

I FLOWCHART (QUICK SORT):

In the following flowchart, the program uses Quick Sort on a single sequence of machine IDs.



SOURCE CODE:

In the following code, the program uses Quick Sort on a single sequence of machine IDs.

FaultyMachine_QuickSorting.py

```python
def quickSort(machines):  3 usages
    if len(machines) <= 1:
        return machines
    else:
        pivot_index = len(machines) - 1  # theIndex of the last element
        pivot = machines[pivot_index]   # theElement at the last index
        left = [x for i, x in enumerate(machines) if x <= pivot and i != pivot_index]#Exclude pivot
        right = [x for x in machines if x > pivot]
        return quickSort(left) + [pivot] + quickSort(right)

def findFaultyMachine(machines):  1 usage
    result = 0
    for machine in machines:
        result ^= machine
    return result

def main():  1 usage
    machines = [303, 101, 404, 202, 101, 303, 404]
    print(f"Original machines: {machines}")
    sorted_machines = quickSort(machines)
    print(f"Sorted machines using Quick Sort: {sorted_machines}")
    faultyMachine = findFaultyMachine(sorted_machines)
    print(f"The faulty machine is: {faultyMachine}")

if __name__ == "__main__":
    main()
```

OUTPUT:

In this program, the machine IDs are sorted in ascending order, then printed as an array.

```
FaultyMachine_QuickSorting ×

"C:\Users\Gian Carlo Trilles\PycharmProjects\pythonProject\.venv\Scripts\python.exe" "C:\Users\Gian
Original machines: [303, 101, 404, 202, 101, 303, 404]
Sorted machines using Quick Sort: [101, 101, 202, 303, 303, 404, 404]
The faulty machine is: 202

Process finished with exit code 0
```